

Cloud & DevOps Engineering

Complete Notes + Interview Q&A

Terraform | Azure | Landing Zone | Migration | Governance | AI Orchestration

Key Mindset

"Code likhna tuccha kaam hai, System Design important hai." — Writing code alone is not enough. Architecture, governance, migration strategy, and security are what truly matter in cloud engineering.

1. Learning Philosophy & Practice Strategy

The right approach to becoming a strong cloud engineer involves structured, repetitive learning with real-world application.

Practice Strategy

- Create separate folders for Resource Groups and Storage Accounts
- Practice each concept at least 10 times manually
- Teach the concept to at least 10 people via screen sharing
- First understand resources manually, then shift to AI-assisted coding
- First 15 days: Build the first 10 Azure resources manually

AI Tools to Use (After Manual Learning)

- Gemini CLI
- Claude Code

Once fundamentals are clear, use AI tools to generate infrastructure faster.

Q: What is the recommended learning approach for Azure?

A: Practice each resource manually at least 10 times, teach it to 10 others via screen sharing, then use AI tools like Gemini CLI or Claude Code to accelerate infrastructure generation. Manual understanding comes first.

Q: Why should you learn manually before using AI tools?

A: Manual practice builds strong foundational understanding of how each Azure resource works. AI tools are productivity multipliers — they only work well when you already know what you're doing.

Q: What mindset should a cloud engineer have?

A: System Design and Architecture thinking is more important than just writing code. A cloud engineer must understand governance, migration, security, and automation — not just syntax.

2. Terraform Topics to Master

Terraform is the primary Infrastructure-as-Code (IaC) tool used for deploying cloud infrastructure in a repeatable, version-controlled way.

Core Terraform Concepts

- Modules — reusable, composable infrastructure units
- Data Block — fetch existing resource information
- Local Block — define reusable local variables
- Dynamic Block — generate repeated nested blocks
- Conditional Expressions — if-else logic in Terraform
- Terraform Functions — built-in string/math/collection functions
- Output Block — expose values after apply
- Generic Modules — modules usable across environments
- Provisioners — run scripts on resources
- custom_data — pass startup scripts to VMs
- Terraform Workspaces — multiple environments, one codebase
- Lifecycle Block — control resource creation/destroy behavior
- Optional Attributes — nullable attributes in variables
- Taint — force resource replacement on next apply
- Multi-subscription Deployment — deploy across Azure subscriptions
- Count Loop in Production — iterate over resources

Common Azure Resources in Terraform

- azurerm_resource_group
- azurerm_virtual_network
- azurerm_virtual_machine
- azurerm_subnet

Folder Structure

Standard Terraform Structure

main.tf — resource definitions provider.tf — provider configuration variables.tf — input variable declarations Environments: nnonprod/ — non-production configs prod/ — production configs

Q: What is a Terraform Module?

A: A module is a reusable container for multiple resources used together. It helps avoid code duplication and enforces consistent configurations across environments.

Q: What is the difference between a Data Block and a Resource Block?

A: A Resource Block creates or manages infrastructure. A Data Block reads/fetches existing infrastructure that was created outside of the current Terraform code — useful for referencing existing VNets, RGs, etc.

Q: What is a Dynamic Block in Terraform?

A: A Dynamic Block allows you to generate repeated nested configuration blocks inside resources (like multiple inbound rules in an NSG) based on a variable list, avoiding code repetition.

Q: What is a Terraform Workspace?

A: Workspaces allow you to manage multiple environments (dev, staging, prod) using the same Terraform code but different state files. Each workspace has an isolated state.

Q: What does the Lifecycle Block do?

A: The Lifecycle Block controls resource behavior: `create_before_destroy` ensures new resource is created before old is deleted; `prevent_destroy` blocks accidental deletion; `ignore_changes` skips specific attribute updates.

3. Cloud Migration Concepts

Migration Types

On-Premises to Cloud

- On-prem → Azure
- On-prem → AWS
- On-prem → GCP

Cloud-to-Cloud Migration

- Azure → AWS, Azure → GCP
- AWS → Azure, AWS → GCP
- GCP → Azure, GCP → AWS

Infrastructure Modernization

- From: Virtual Machines
- To: AKS, ACS, OpenShift, Serverless, Web Apps

The 7 Rs Migration Strategies

- Rehost (Lift & Shift) — move as-is to cloud
- Replatform — move with minor optimizations (e.g., managed DB)
- Refactor / Re-architect — redesign for cloud-native
- Repurchase — replace with SaaS product
- Retire — decommission unused apps
- Retain — keep on-prem for now
- Relocate — move VMs to cloud without OS change

Q: What is Lift and Shift migration?

A: Lift and Shift (Rehost) is moving an application from on-premises to the cloud without making any code or architecture changes. It's the fastest migration method but does not leverage cloud-native features.

Q: What is the difference between Replatform and Refactor?

A: Replatform makes minor changes to benefit from cloud (e.g., moving to a managed database), while Refactor involves re-architecting the application to be cloud-native — often breaking it into microservices or using containers.

Q: What is the difference between Brownfield and Greenfield deployment?

A: Brownfield refers to migrating or working with existing applications/infrastructure already running on-premises. Greenfield means building completely new applications and infrastructure from scratch on the cloud.

4. Azure Landing Zone

An Azure Landing Zone is the foundational architecture and governance baseline for enterprise cloud deployments.

What is a Landing Zone?

- A structured, pre-configured cloud foundation
- A governed environment enforcing security and compliance
- A scalable deployment baseline for enterprise workloads
- An environment covering: Networking, Identity, Security, Monitoring, and Governance

Why Landing Zone?

- Enterprise environments require standardized governance
- Networking and identity must be consistent across teams
- Security and compliance must be enforced at scale
- Prevents sprawl and ensures auditability

Hierarchy Models

Simple Hierarchy

- Management Group → Subscription → Resource Group → Resources

Hub and Spoke Architecture

- Hub VNet — hosts shared services: Firewall, VPN Gateway, DNS
- Spoke VNets — host individual application workloads
- Spokes connect to Hub via VNet Peering
- Benefits: centralized security, easier management, traffic control

Q: What is an Azure Landing Zone?

A: An Azure Landing Zone is a pre-configured, governed cloud environment that provides the foundational setup needed for enterprise workloads — including networking, identity, security policies, and compliance controls.

Q: What is Hub and Spoke architecture?

A: Hub and Spoke is a networking model where a central Hub VNet hosts shared services (firewall, VPN, DNS) and multiple Spoke VNets host applications. Spokes connect to the Hub via VNet Peering — traffic is centrally inspected and governed.

Q: What is the Azure Management Group hierarchy?

A: Root Management Group → Management Groups (departments/teams) → Subscriptions → Resource Groups → Resources. Policies applied at higher levels are inherited by all children.

5. Governance & Compliance

Azure Policy

- Built-in Policies — pre-defined rules (e.g., require HTTPS, enforce tagging)
- Custom Policies — org-specific rules written in JSON
- Initiatives (Policy Sets) — groups of policies applied together

Compliance Areas

- Security standards (CIS, ISO 27001, NIST)
- Resource tagging enforcement
- Region restrictions (lock resources to specific regions)
- Naming conventions

Q: What is Azure Policy?

A: Azure Policy is a governance service that allows you to create, assign, and manage rules (policies) that enforce standards across your Azure resources. Policies can audit or deny non-compliant deployments automatically.

Q: What is the difference between a Policy and an Initiative?

A: A Policy is a single rule (e.g., 'all resources must have a CostCenter tag'). An Initiative is a collection of multiple related policies bundled together for easier assignment — e.g., a 'Security Baseline' initiative containing 20 security policies.

Q: How do you enforce resource tagging in Azure?

A: Use Azure Policy to require specific tags on all resources or resource groups. You can deny the creation of resources without required tags, or auto-remediate by adding missing tags using a DeployIfNotExists or Modify policy effect.

6. Identity & Access Management (IAM)

Key IAM Components

- Entra ID (formerly Azure AD) — cloud identity provider
- Users & Groups — organize identities
- RBAC (Role-Based Access Control) — assign permissions by role
- MFA (Multi-Factor Authentication) — second factor for login
- PIM (Privileged Identity Management) — just-in-time elevated access
- App Registrations — register applications for OAuth/API access
- AD Sync — sync on-premises Active Directory to Entra ID
- On-Prem AD Integration — hybrid identity

RBAC Roles

- Owner — full access including delegation
- Contributor — manage resources, no access control
- Reader — read-only access
- Custom Roles — organization-defined permission sets

Q: What is RBAC in Azure?

A: Role-Based Access Control (RBAC) assigns permissions to users, groups, or service principals based on roles (Owner, Contributor, Reader or custom roles) scoped to management groups, subscriptions, resource groups, or individual resources.

Q: What is PIM (Privileged Identity Management)?

A: PIM provides just-in-time privileged access — users request elevated roles only when needed for a limited time, require approval, and all activity is audited. This reduces standing privileged access and improves security.

Q: What is Managed Identity?

A: Managed Identity is an Azure feature that gives Azure resources (VMs, App Services, etc.) an automatically managed identity in Entra ID. Resources can authenticate to other Azure services (like Key Vault, Storage) without storing credentials in code.

7. Networking Architecture

Core Networking Components

- VNet (Virtual Network) — isolated network in Azure
- Subnet — segment of a VNet for resource placement
- VNet Peering — direct connection between VNets (low latency, no gateway)
- VPN Gateway — encrypted connection to on-premises or other clouds
- ExpressRoute — dedicated private circuit from on-prem to Azure

Advanced Networking Services

- Azure Load Balancer — Layer 4, distributes TCP/UDP traffic
- Application Gateway — Layer 7, HTTP-aware with WAF support
- Azure Front Door — global CDN + load balancing + WAF
- Traffic Manager — DNS-based global traffic routing

Q: What is the difference between VNet Peering and VPN Gateway?

A: VNet Peering directly connects two VNets over Microsoft's backbone — low latency, no bandwidth limit. VPN Gateway creates an encrypted IPsec tunnel — used for on-premises connectivity or cross-region connections through the internet.

Q: What is the difference between Load Balancer and Application Gateway?

A: Azure Load Balancer operates at Layer 4 (TCP/UDP) and routes traffic based on IP and port. Application Gateway operates at Layer 7 (HTTP/HTTPS) and can route based on URL path, host header, and provides WAF (Web Application Firewall) capabilities.

Q: What is ExpressRoute?

A: ExpressRoute provides a private, dedicated connection from your on-premises data center to Azure — bypassing the public internet. It offers consistent performance, higher security, and SLAs for enterprise hybrid connectivity.

8. Security Architecture

Key Security Components

- Application Gateway with WAF — protect web apps from OWASP attacks
- Azure Firewall — stateful, centralized network traffic filtering
- NSG (Network Security Group) — subnet/NIC level inbound/outbound rules
- ASG (Application Security Group) — group VMs for NSG rules
- Azure Bastion — secure browser-based RDP/SSH without public IPs
- Key Vault — store secrets, keys, certificates securely
- Private Endpoints — access Azure services over private IP
- Microsoft Sentinel — SIEM and SOAR for threat detection
- Microsoft Defender for Cloud — posture management and threat protection
- Managed Identity — passwordless authentication for Azure resources
- DDoS Protection — network-level volumetric attack mitigation

Zero Trust Model

- Never trust, always verify — no implicit trust based on network location
- Verify identity continuously — MFA, conditional access policies
- Restrict lateral movement — micro-segmentation, least privilege

Q: What is the Zero Trust security model?

A: Zero Trust assumes no user or resource is trusted by default — even inside the network. Every access request must be explicitly verified through identity validation, device compliance, and least-privilege principles. It limits the blast radius of breaches.

Q: What is the difference between NSG and Azure Firewall?

A: NSG is a basic stateful packet filter at subnet or NIC level — allows/denies traffic based on IP, port, and protocol. Azure Firewall is a centralized, fully stateful managed firewall with FQDN filtering, threat intelligence, and centralized logging — used at the hub level in Hub-Spoke topologies.

Q: What is Azure Key Vault?

A: Azure Key Vault is a managed service for securely storing and accessing secrets (passwords, API keys), cryptographic keys, and certificates. Applications access secrets at runtime without hardcoding credentials — integrated with Managed Identity for passwordless access.

9. Compute Services

Azure Compute Options

- Virtual Machines (VM) — IaaS, full control over OS and software
- VM Scale Sets (VMSS) — auto-scale group of identical VMs
- Web Apps / App Services — PaaS for web and API hosting
- Azure Functions — serverless, event-driven compute
- ACS (Azure Container Service) — container orchestration
- AKS (Azure Kubernetes Service) — managed Kubernetes cluster

Q: What is the difference between VM and AKS?

A: A VM is IaaS — you manage the OS, patching, and scaling. AKS is a managed Kubernetes service — you deploy containerized applications and Azure manages the control plane. AKS is better for microservices and container workloads.

Q: When should you use Azure Functions over App Services?

A: Use Azure Functions for event-driven, short-duration tasks (processing queue messages, HTTP triggers, timers) where you pay per execution. Use App Services for long-running web applications and APIs where you need persistent compute with more control.

Q: What is VM Scale Sets (VMSS)?

A: VMSS allows you to deploy and manage a group of identical, auto-scaling VMs. When load increases, it automatically adds VMs; when load drops, it removes them. Ideal for stateless workloads like web tier servers.

10. Database & Storage

Database Services

- Azure SQL / SQL Server — relational, enterprise-grade
- PostgreSQL Flexible Server — open-source relational DB
- MySQL Flexible Server — popular for web apps
- CosmosDB — globally distributed, multi-model NoSQL database

Storage Services

- Storage Account — foundational storage service
- Blob Storage — unstructured object storage (images, backups, logs)
- File Share — SMB/NFS shared file storage for VMs

Data Strategy

- Backup — automated backup policies for DB and VM
- Replication — geo-redundant replication for HA
- DR Strategy — active-active or active-passive failover

Q: What is CosmosDB?

A: CosmosDB is a globally distributed, multi-model NoSQL database supporting multiple APIs (SQL, MongoDB, Cassandra, etc.). It offers single-digit millisecond latency, 99.999% availability SLAs, and automatic global replication — ideal for globally distributed applications.

Q: What is the difference between Blob Storage and File Share?

A: Blob Storage stores unstructured data as objects accessed via HTTP/REST — used for backups, images, logs, and static files. Azure File Share provides SMB/NFS network file storage — used to replace traditional file servers or mount shared drives to multiple VMs.

11. Monitoring & Observability

Monitoring Stack

- Azure Monitor — centralized monitoring platform for all Azure resources
- Log Analytics Workspace — stores and queries logs using KQL
- Application Insights — APM for application performance and traces
- Prometheus — metrics collection for Kubernetes workloads
- Grafana — visualization dashboards for metrics

Q: What is the difference between Azure Monitor and Log Analytics?

A: Azure Monitor is the umbrella platform that collects metrics and logs from all Azure resources. Log Analytics Workspace is where those logs are stored and queried using KQL (Kusto Query Language). Log Analytics is a component within Azure Monitor.

Q: What is Application Insights?

A: Application Insights is an Application Performance Monitoring (APM) service. It tracks request rates, response times, failure rates, dependency calls, and traces for web applications — helping identify performance bottlenecks and errors in real-time.

12. Backup & Disaster Recovery

Backup Types

- VM Backup — Azure Backup for full/incremental VM snapshots
- Database Backup — automated backups for SQL, PostgreSQL, MySQL
- Storage Account Backup — blob-level backup and soft delete

DR Concepts

- RTO (Recovery Time Objective) — max acceptable downtime after disaster
- RPO (Recovery Point Objective) — max acceptable data loss (time)
- Active-Active — both regions serving traffic simultaneously
- Active-Passive — primary region active, secondary on standby
- Failover Testing — regular drills to validate recovery procedures

Q: What is the difference between RTO and RPO?

A: RTO (Recovery Time Objective) is the maximum time your business can tolerate being offline after a disaster. RPO (Recovery Point Objective) is the maximum amount of data loss (measured in time) acceptable — e.g., RPO of 1 hour means backups every hour.

Q: What is the difference between Active-Active and Active-Passive DR?

A: Active-Active: both regions serve live traffic simultaneously — zero failover time, higher cost. Active-Passive: secondary region is on standby, serving no traffic until primary fails — lower cost but some failover delay depending on RTO requirements.

13. High Availability

HA Concepts

- Availability Sets — distribute VMs across Fault and Update Domains within a datacenter
- Availability Zones — physically separate datacenters in the same Azure region
- Load Balancing — distribute traffic across healthy instances
- Multi-Region Architecture — deploy across regions for geographic redundancy

Q: What is the difference between Availability Sets and Availability Zones?

A: Availability Sets distribute VMs across Fault Domains (separate racks/power) and Update Domains (separate maintenance windows) within a single datacenter — protecting against hardware failures. Availability Zones are separate physical datacenters within a region — protecting against full datacenter failures.

Q: What SLA do Availability Zones provide?

A: Azure guarantees 99.99% uptime SLA when VMs are deployed across two or more Availability Zones in the same region, compared to 99.95% for Availability Sets.

14. Migration Phases

Phase 1 — Discovery & Assessment

Business Discovery

- Identify business owners and application criticality
- Assess downtime tolerance and SLAs
- Document developer stack, databases, and dependencies
- Analyze traffic patterns and peak usage

Infrastructure Discovery

- Inventory VMware/Hypervisor workloads
- Document IP address spaces and CIDR ranges
- Map firewalls, routing, and DNS configurations
- List certificates, domains, and backup/DR setup

Phase 2 — Migration Strategy

- Select appropriate migration strategy from the 7 Rs
- Prioritize applications by complexity and criticality

Phase 3 — Landing Zone Design

- Design Governance, Networking, Security, Identity, and Monitoring

Phase 4 — Implementation

- Write infrastructure as code using Terraform
- Push code to source control (SCM)
- Deploy via CI/CD pipelines (Azure DevOps / GitHub Actions)

Q: What happens in the Discovery & Assessment phase?

A: The team conducts business discovery (owners, SLAs, downtime tolerance, tech stack) and infrastructure discovery (VMs, networking, firewalls, certificates, backup). This phase produces the inventory and dependency map needed to plan the migration.

Q: Why is Landing Zone design a separate phase?

A: Before migrating workloads, the target cloud environment must be properly set up with governance policies, networking topology, security controls, and identity management. Without a Landing Zone, workloads land in an ungoverned environment causing security and compliance issues.

15. Documentation

Required Documents

- HLD (High Level Design) — architecture overview, component diagram
- LLD (Low Level Design) — detailed resource specs, IP ranges, configs
- Networking Diagram — visual representation of VNets, subnets, peering
- Firewall Documentation — firewall rules and traffic policies
- NSG Rules — inbound/outbound rules for each subnet/NIC
- Operational Readiness — runbooks, escalation paths, contacts
- Runbooks — step-by-step operational procedures

Q: What is the difference between HLD and LLD?

A: HLD (High Level Design) provides a high-level architecture overview showing major components and their relationships — suitable for stakeholders. LLD (Low Level Design) provides detailed technical specifications — IP ranges, resource names, NSG rules, config values — used by engineers during implementation.

16. AI Orchestrator Architecture

The final architecture in the diagram describes an AI-driven infrastructure orchestration system combining agents, MCP integrations, and RAG pipelines.

Agent Types

- Kubernetes Agent — manages K8s cluster operations
- Azure DevOps Agent — triggers and monitors CI/CD pipelines
- GitHub Agent — manages source control, PRs, branches
- Azure Agent — provisions and manages Azure resources
- RAG Agent — handles document retrieval and context injection

MCP Integration Layer

- K8s MCP — Kubernetes Model Context Protocol integration
- ADO MCP — Azure DevOps integration
- GitHub MCP — GitHub integration
- Azure MCP — Azure resource management integration

RAG Pipeline

- Documents → Chunking → Embeddings → Vector DB
- RAG Agent queries Vector DB for contextual knowledge

Decision Engine

- Orchestration layer that selects tools and connects agents
- Processes multi-step workflows across systems

Q: What is a RAG (Retrieval-Augmented Generation) pipeline?

A: RAG is an AI architecture pattern where documents are chunked, converted into vector embeddings, and stored in a Vector DB. When a query is made, relevant document chunks are retrieved and injected as context into the AI model's prompt — enabling accurate, knowledge-grounded responses.

Q: What is MCP (Model Context Protocol)?

A: MCP is a protocol that allows AI agents to connect to and interact with external tools and services (like Kubernetes, Azure DevOps, GitHub) in a standardized way — enabling the AI model to take actions in those systems as part of an automated workflow.

Q: What is an AI Orchestrator?

A: An AI Orchestrator is a decision-making layer that coordinates multiple AI agents, tools, and workflows. It selects which agent/tool to invoke based on the task, manages context between steps, and executes multi-step automation pipelines across infrastructure systems.

17. Quick Reference Summary

Key Terms Cheat Sheet

Term	Definition
IaC	Infrastructure as Code — define infrastructure using code files (Terraform, ARM)
Landing Zone	Pre-configured, governed Azure environment for enterprise workloads
Hub-Spoke	Networking topology: central Hub + connected Spoke VNets
RBAC	Role-Based Access Control — permissions assigned by role
PIM	Privileged Identity Management — just-in-time elevated access
NSG	Network Security Group — subnet/NIC level traffic rules
WAF	Web Application Firewall — protects against OWASP Top 10
Bastion	Secure browser-based RDP/SSH — no public IP needed on VMs
Key Vault	Managed secrets, keys, certificates storage
VMSS	VM Scale Sets — auto-scaling group of identical VMs
AKS	Azure Kubernetes Service — managed Kubernetes
RTO	Recovery Time Objective — max tolerable downtime
RPO	Recovery Point Objective — max tolerable data loss
RAG	Retrieval-Augmented Generation — AI with document context
MCP	Model Context Protocol — AI agent tool integration standard
VNet Peering	Direct low-latency connection between two Azure VNets
ExpressRoute	Private dedicated connection from on-prem to Azure
Sentinel	Azure SIEM/SOAR — security event detection and response
Entra ID	Azure Active Directory — cloud identity provider
HLD/LLD	High Level Design / Low Level Design documents

— End of Notes —